

Classical Macroeconomics: Classical Monetary and Employment Theory

The macroeconomic theories now must widely be accepted among economists constitute a marriage of two quite diverse approaches: one commonly called **Keynesian**, the other **Classical** (1776-1930s): the former is the product of the 1930s and the latter is an accumulation and refinement of ideas developed during the previous century and a half. Such a marriage of Keynesian and classical elements- sometimes described as *Keynesian-classical synthesis* (Samuelson termed it the neoclassical synthesis model). Nevertheless, for purposes of presentation of this synthesis, it has been found useful to present separately, and in its starkest form, the basic elements of the classical theory without any Keynesian admixture.

The main body of classical economics was microeconomic in character. It dealt with the determination of (relative) prices for and quantities of individual goods and services, the determination of (relative) incomes to individual factors of production, and the determination of (relative) prices of classes of assets, real and financial. Its principal explicit macroeconomic subject was the determination of the money price level, which, however, was believed to have essentially no significance for relative prices and incomes.

The reconstructed classical macroeconomics stands on three pillars:

1. Say's Law
2. The Quantity Theory of Money
3. Perfect Wage-Price Flexibility

1. Say's Law

One of the few explicitly macroeconomic propositions of classical theory was Say's Law of Markets (named after J.B. Say, the French economist, 1767-1832). Say's law is usually summarized as "Supply creates its own demand". If goods are produced, there will automatically be a market for them. This can be easily be seen to be true in a barter economy, although its application was not supposed to be limited to that circumstance. What Say appears to have had in mind can be expressed in this way: people work not for its own shake but only to obtain goods and services that yield satisfactions. The very act of production constitutes the demand for other goods, a demand equivalent to the value of the surplus goods each man produces. How can there ever be a general overproduction of goods? Each man's production (supply) constitutes his demand for other goods; hence the AD must in some sense equal to AS. What Say's law implies is that any increment of output will generate an equivalent

increase in income and in spending. Thus, income and product can always be at a 'full-employment' level.

When Say's law is framed in terms of a barter economy, it seems self-evident. But does it also true for an economy using money? Can we be sure that the money received will be promptly and fully spent against other goods? We shall see latter while discussing the monetary theory contemporary with Say's law did explain how and why money's role was solely that of a 'veil', behind which everything operated in the same way as if all exchanges took the form of barter. But this result depends on many specific assumptions about behavior, which might be subject to challenge.

The elementary formulation of Say's law also paid no attention to the phenomenon of saving and investment. Thus, to understand the entire basis for Say's law in a real-world economy- which was money, not barter; in which most people sell labor services, not final product; and in which people save and invest- we need to look at a whole range of classical ideas. The first of these basic classical ideas is the "quantity theory of money".

2. The Quantity Theory of Money

The quantity theory asserts that money determines only the price level, not real output. The root idea of the quantity theory is that no rational person holds money idle, for it produces nothing and yields no satisfaction. Rather, people promptly use all the cash they receive from the sale their goods and services to buy other goods and services. How promptly that occurs depends on how production is organized, how frequently incomes are paid, and other structural or institutional factors that were judged to be quite independent of the quantity of money or the level of prices. Given these assumptions, the theory showed how the quantity of money determines the level of money prices, and in a way which is without impact on the demand or supply of any individual product, and thus cannot affect relative prices nor the absolute quantities of products produced and sold. This, it cannot affect aggregate production, either.

The quantity theory of money was discussed by the famous 18th century philosopher David Hume and has been advocated more recently by the prominent economist Milton Friedman. Most economists rely on this theory for explaining the long-run determinants of the price level and the inflation rate. The value of money (like any other commodity) is determined by the supply and demand for money. Formally, we can state the quantity theory in the following equation:

$$M \times V_T = P \times T \dots \dots \dots (1)$$

Where T is the number of times in a year that goods and services are exchanged for money. P is the price of a typical transaction- the number of taka exchanged. The product $P \times T$ equals the amount

of taka exchanged in a year. M is the quantity of money in circulation, V_T is the transaction velocity of money and measures the rate at which money circulates in the economy. In other words, V_T tells us the number of times a taka note changes hands in a given period of time.

For example, suppose that 60 sandwiches are sold in a given year at \$50 per sandwich. Then T equals 60 per year and P equals \$50 per sandwich. Therefore,

$$P \times T = 50 \times 60 = \$300/\text{year}$$

Now, suppose the quantity of money (M) in the economy is \$100 in one taka bill. Then,

$$V_T = PT/M = 300/100 = 3 \text{ (times per year)}$$

The quantity equation is an identity: the definitions of the four variables make it true.

From Transaction (T) to Income (real GDP):

Transactions (T) and output (Y) are closely related, because the more the economy produces, the more goods and services are bought and sold. They are not the same, however, when one person sells a used car to another person, they make a transaction using M , even though the used car is not part of current output. If Y denotes the amount of output, P denotes the price of one unit of output, then nominal value of output is PY , where Y is real GDP and P is the GDP deflator; then the quantity equation becomes-

$$M \times V_Y = P \times Y \dots \dots (2)$$

Where V_Y now the income velocity of money. This version of the quantity of money equation is the most common, and it is the one we use from now on.

Velocity and the Demand for Money

We now define,

$$M_d = mPY$$

Where M_d is the demand for money, and m is that (constant) fraction of PY which, in the aggregate, all the people in the economy desire to hold in the form of money balances; and

$$M = M_d$$

$$M = mPY \dots \dots (3)$$

Where M is the supply of money. Equation (3) says that all the money that exists must be in

someone's balances, and, in equilibrium, it must be willingly hold. It should be obvious that $m = \frac{1}{V_Y}$.

It is therefore clear that whatever factors determined the sign of V_Y must also determine the sign of m . However, whereas the formulation in terms of V_Y directed attention towards the institutional and

structural factors which influences the frequency of money payments- which operates as constraints on behavior- the formulation in terms of m directs attention to the factors on attitudes which influence people to hold money. Both sets of factors are obviously important in thinking about the phenomenon involved. Formulation (3) makes it easier to understand that the quantity theory equation expresses an equilibrium condition.

3. Perfect Wage-Price flexibility: Employment & Production

We saw earlier that classical macroeconomics maintained that $P \times Y$ is determined by the money supply M . With a given stock of money, aggregate price times real GDP is a constant. More can be sold only at proportionally lower prices. If prices are flexible, they will fall whenever there are idle resources; this fall in prices somehow automatically absorb these idle resources and expand economy's output. Here we examine how price flexibility promotes full employment in an economy in which people work for wages, rather than supply final goods directly to the market.

Aggregate Supply of Labor

The total supply of labor hours was assumed by classical economists to vary directly with the real hourly wage that potential workers found they could earn- that is, the more workers would seek jobs the higher the available real wages. However, at least at relatively high real wages, the supply might be argued to depend inversely on the real wage- even on classical assumptions. For already well-paid workers might wish to take some part of the higher real income made possible by a still higher real wage in the form of more leisure.

Denoting the aggregate supply L as a function of the real wage- $\frac{W}{P}$ (the ratio of the money wage W to the price level P), we have

$$L = L\left(\frac{W}{P}\right) \dots \dots \dots (4)$$

Where the sign of the first derivative of this function is unknown.

$$\frac{dL}{d(W/P)} > < 0$$

Aggregate Demand for Labor and Labor Market Equilibrium

For the purposes of classical macroeconomics, it was deemed necessary to establish the total demand for labor by all firms as a function of the economy-wide real wage, i.e., of $\frac{W}{P}$ where P is some average of all of the individual P_j . The usual way in which this is done is to treat the entire economy analogously with a single firm. Thus, we might suppose that there exists an aggregate production function,

$$Y = F(N, K) \dots \dots \dots (5)$$

Where Y is total output, N is total employment, and K is the aggregate stock of capital. Then, we have for maximum profit:

$$\begin{aligned} \text{profit, } \pi &= PY - WN - rk \\ &= PF(N, K) - WN - rk \end{aligned}$$

$$\therefore \frac{\partial \pi}{\partial N} = P \frac{\partial F}{\partial N} - W = 0$$

$$\text{or, } \frac{\partial F}{\partial N} = \frac{W}{P} \quad (q) \dots \dots \dots (5a)$$

It says that the aggregate marginal product should equal the economy-wide real wage.

This system is completed by the prescription that equilibrium requires full employment:

$$L = N \dots \dots \dots (6)$$

But we must admit that the above procedure is highly unsatisfactory if not incorrect.

There are margin difficulties of aggregating nonlinear well-behaved production functions of the individual firms. Even if it exists, aggregate data fails to discover any evidence of diminishing returns. If we might wish to ignore these difficulties and to assume that aggregate production function (5) is well behaved, then it is unclear what the condition (5a) is supposed to represent. What profit maximization condition does it here refer to: the aggregate of all firms' profits? There seems little point in defining such an aggregate profit, since no individual or government decision is based on it. Certainly there is no decision to maximize it.

Now we may represent the "traditional version" of the classical system by equation (4), (5) (5a) and (6) as follows:

$$\begin{aligned} L &= L(W/P); \quad \frac{dL}{d(W/P)} > < 0 \\ Y &= F(N, K); \quad \frac{\partial F}{\partial N} > 0; \quad \frac{\partial}{\partial N} \left(\frac{\partial F}{\partial N} \right) < 0 \\ \frac{\partial F}{\partial N} &= \frac{W}{P} \\ L &= N \end{aligned}$$

However, the system involves four macro variables: Y, P, W, N & L, but only four equations. To make the system determinate, we explicitly add the quantity theory equation (3):

$$M = mPY$$

The quantity theory provides a classical answer to the question what happens to the general price level, P as W falls in the presence of unemployment. So long as $L > N$, W falls without limit. Now, however, there is a lower limit to the fall of P ; it is the point at which $M = mPY_p$, where Y_p is the potential output. This is the level of output found from the aggregate production-employment relationship (5), and corresponds to the N at which $N = L$. Given the real wage W/p , which is necessary for $N = L$, and given the P that satisfies the quantity theory when $Y = Y_p$, the necessary W can be determined. If W falls without limit so long as $L > N$, this equilibrium will be established and maintained.

A graphical Representation of this model is given below in Figure-1:

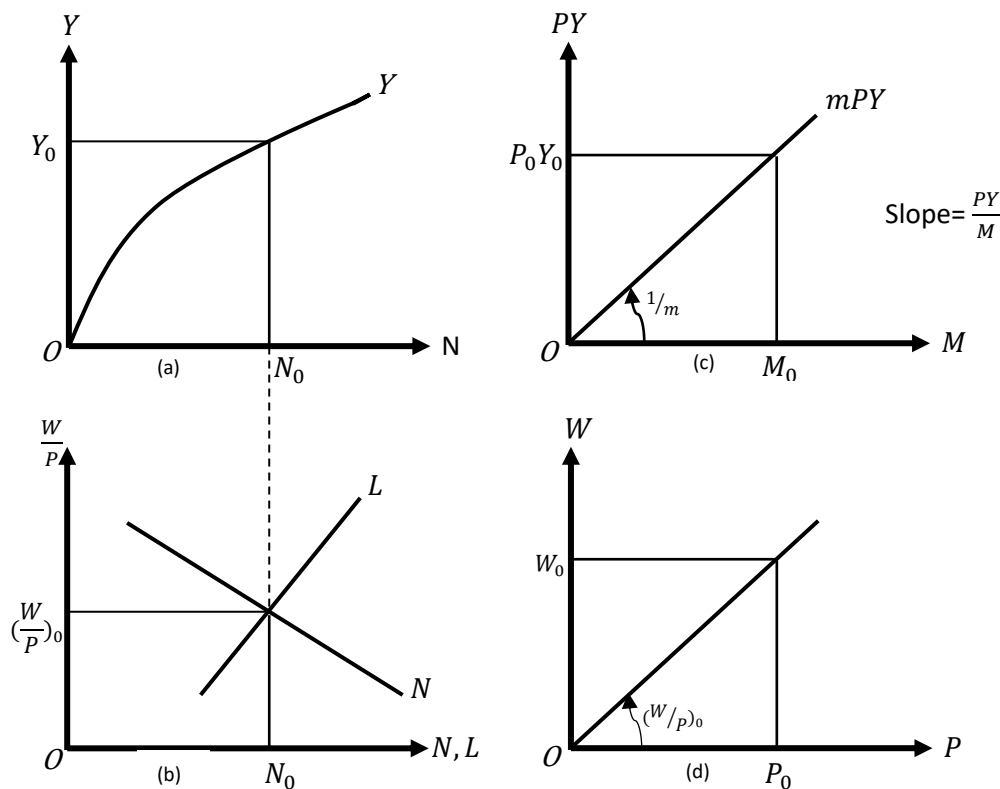


Figure-1: Classical Macro Model (Reconstructed)

Part (a) shows the aggregate production function subject to diminishing returns. Part (b) shows the intersection of supply and demand curves for labors. The curve N represents the slope of the production function of part (a), which declines as employment increases. A change in the height of the production function, with no change in its slope at any level of employment, would leave the

marginal product curve unchanged. But any change in the slope of the production function will alter the marginal product, i.e., the N -curve.

The intersection of the two curves in part (b) defines the point of full employment, N_0 and the real wage, $(\frac{W}{P})_0$, which corresponds to full employment. Assuming perfect labor market competition, the situation- $L > N$ implies a rapid decline in W and for $N > L$ implies a rapid rise in W . Stability of the W is obviously a condition of equilibrium of the system. Therefore, equilibrium requires stable W and a stable P which produces a real wage of $(\frac{W}{P})_0$.

We find the equilibrium P_0 in point (c) of the figure. Here the straight line through the origin, mPY with slope $\frac{1}{M}$ shows the amount of money (M) required for each level of money income (PY) or read in reverse, the level of PY which each possible quantity of M can support. If the actual stock of money is M_0 , then money income must equal $(PY)_0$. Since we know the equilibrium income Y_0 from part (a), we can immediately complete the equilibrium P_0 .

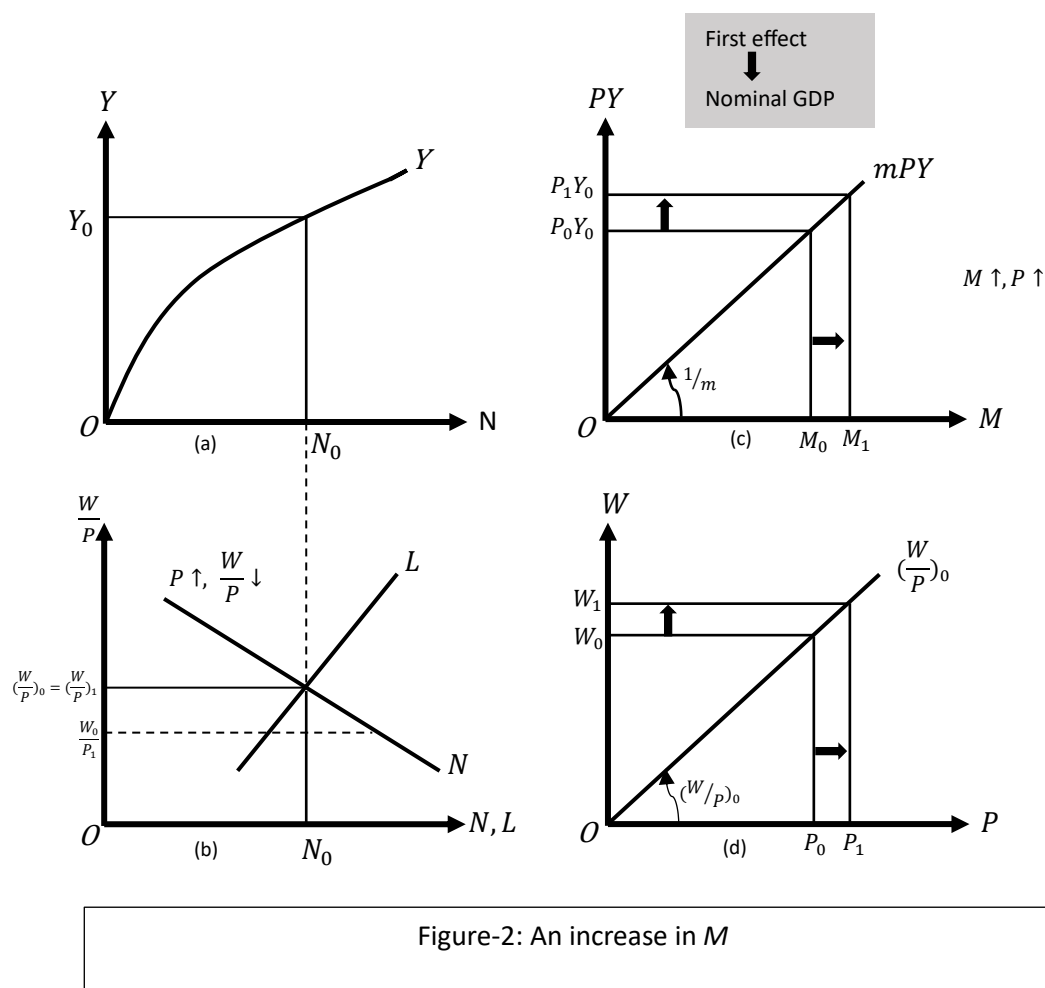
Part (d) permits us to find the necessary level of nominal wage, W . Given the equilibrium $(\frac{W}{P})_0$ and the equilibrium P_0 , there is only one $W = W_0$ consistent with both of these. This is read off in part (d) as the vertical coordinate of the intersection of equilibrium P_0 and the equilibrium $(\frac{W}{P})_0$ angle.

Comparative Static Analysis

A better understanding of the model will come from analyzing the effects on equilibrium values of several variables of various changes in parameters. We may consider the following two cases:

- i. The effects of an increase in M .
- ii. A shift in the production function.

Case (i):



An increase in M permits an increase in the product of P and Y as shown in part (c) of figure-2. The previous output Y_0 , could now sell at a higher price, P_1 . If W did not rise, real wage $\frac{W_0}{P_1}$ would fall. This would induce employers to try to increase output, bidding against each other for workers. Since no more workers are to be had, the money wage, W , must rise far enough to eliminate the excess demand. The result of an increase in M , then, is to raise wages and prices in equal proportion, leaving output (Y), real wages ($\frac{W}{P}$), and employment N unaffected. The result of a decrease in M can easily be worked out.

Case (ii):

We shall consider a change in the production function. The change shown in figure-3 involves an increase in both the average and the marginal product of labor.

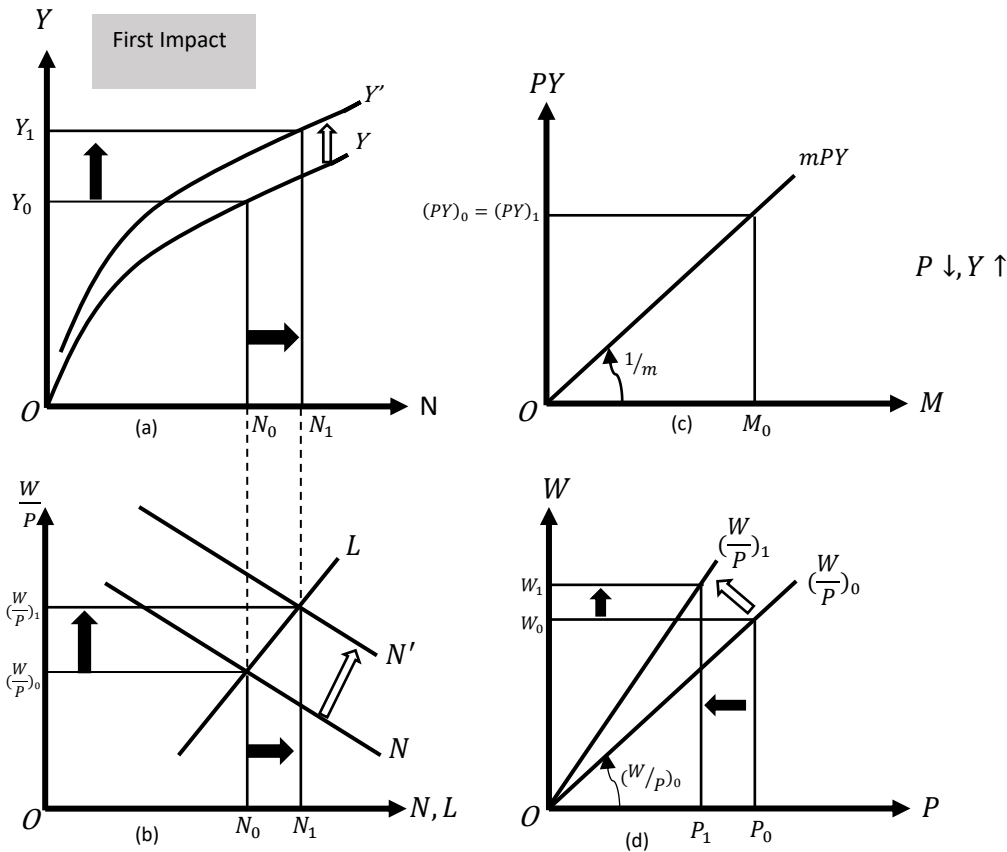


Figure-3: A shift in the Production Function

The new production function in figure-3, Y' , gives a new marginal product curve, or demand-for-labor curve, N' . The equilibrium real wage is increased from $(\frac{W}{P})_0$ to $(\frac{W}{P})_1$, as is the equilibrium volume of employment, from N_0 to N_1 . Output is enlarged to Y_1 , both because of the higher production function and because of the larger employment. If M and m are unchanged, the larger output Y_1 , can be sold only at a lower price level, P_1 . Despite the drop in prices, the money wage (W) can nevertheless rise to W_1 . Depending on the slopes of the various functions, the new money wage, W_1 , might be lower rather than higher; however, the real wage $(\frac{W}{P})_1$ will certainly be higher- prices will fall more than wages.

We can derive graphically the effect of an increase in the supply of labor, L . They are: a lower real wage, a larger employment, a larger output, a lower price level and a lower money wage.

Note that, we have divided the above analysis into two parts- a "real" and a "monetary". The real wage and the level of employment and output are determined by "real" factors alone; the marginal productivity of labor and marginal disutility of labor. Money wages and prices are determined solely by monetary factors. Changes in the "real" side factors can affect prices and wages, but changes on

the monetary side have no effect on the “real” magnitudes. This separation of variables into these two parts is called the classical dichotomy. This presumably justifies the classical tradition of developing the theory of output and employment entirely in real terms (Say’s law), leaving the theory of absolute prices to a separate treatise on monetary theory.

The essential conclusions of the “classical model” are valid so long as we assume perfect wage-price flexibility, and money demanded only as a medium of exchanges.

NB: This makes it clear that the conclusions of the classical model do not rest on the existence of an aggregate production function, a diminishing marginal product of labor, profit maximization, nor the existence of perfect competition in all product markets.

Rigid Wages and Monetary Policy in the Classical Model:

We consider now the effects of an absence of any tendency for money wages to fall so long as men are unemployed. This might result from labor unions or government wage policy. Any of these can create unemployment if the money wage (W) is held or raised too high.

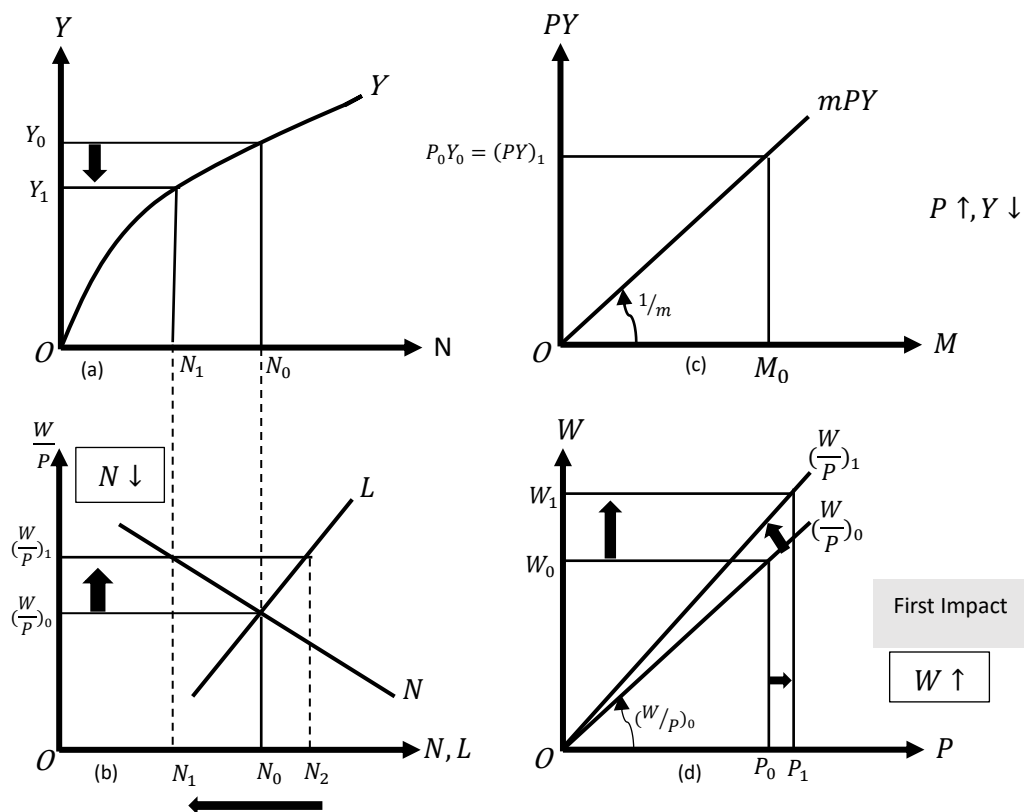


Figure-4

Let us consider the situation in figure-4, with strictly classical assumptions in all respects other than money wages (W) flexibility. Suppose that the money wage was originally W_0 , but was pushed to W_1 . What will be the new level of P , N , Y and $\frac{W}{P}$?

In the first place, we can easily see that P must rise, at least somewhat. For if P did not rise, $\frac{W}{P}$ would be higher than before, and employers would produce less; and a smaller Y with no increase in P would be inconsistent with a constant M & m . P must rise.

Second, we can see that prices (P) could not rise as much as did W . For if P rise in the same proportions, $\frac{W}{P}$ would be unchanged; employers would want to produce as much as before. However, for them to sell the same Y , at a higher P , is also inconsistent with constant M and m . The new PY must be the same as before; therefore, P must rise and Y must fall.

The new equilibrium, as shown in figure-4, must involve both a higher P and higher $\frac{W}{P}$ than before; therefore, it necessarily involves less employment (N) and Y . The results of an arbitrary increase in W thus are higher P , higher $\frac{W}{P}$, smaller N and Y and a volume of unemployment equal to $N_1 - N_2$. Those workers still employed are clearly better off.

Clearly, Monetary policy could be used to offset the effects of too high a W . If unemployment develops because the W fails to decline, monetary expansions could permit full-employment level to be restored. In this particular situation, an expansion in M need not in any meaningful sense be inflationary. P would merely have to be prevented from falling in response to larger output (Y).

The classical analytical tools have weaknesses which make them not fully suitable for application to public policy. Of their logical completeness and consistency, however, there can be no doubt. This reconstructed "Classical theory" cannot be attacked except by attacking the behavioral or technical assumptions on which it is based. Keynes was surely wrong in sometimes attacking it as logically incomplete as inconsistent.